INCH-POUND

MIL-DTL-25708/1E
23 July 1999
SUPERSEDING
MIL-A-25708/1D
28 May 1987

DETAIL SPECIFICATION SHEET

ANTENNA, BLADE, AT-741B/B (HN CONNECTOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the antenna described herein shall consist of this document and the latest issue of MIL-DTL-25708.

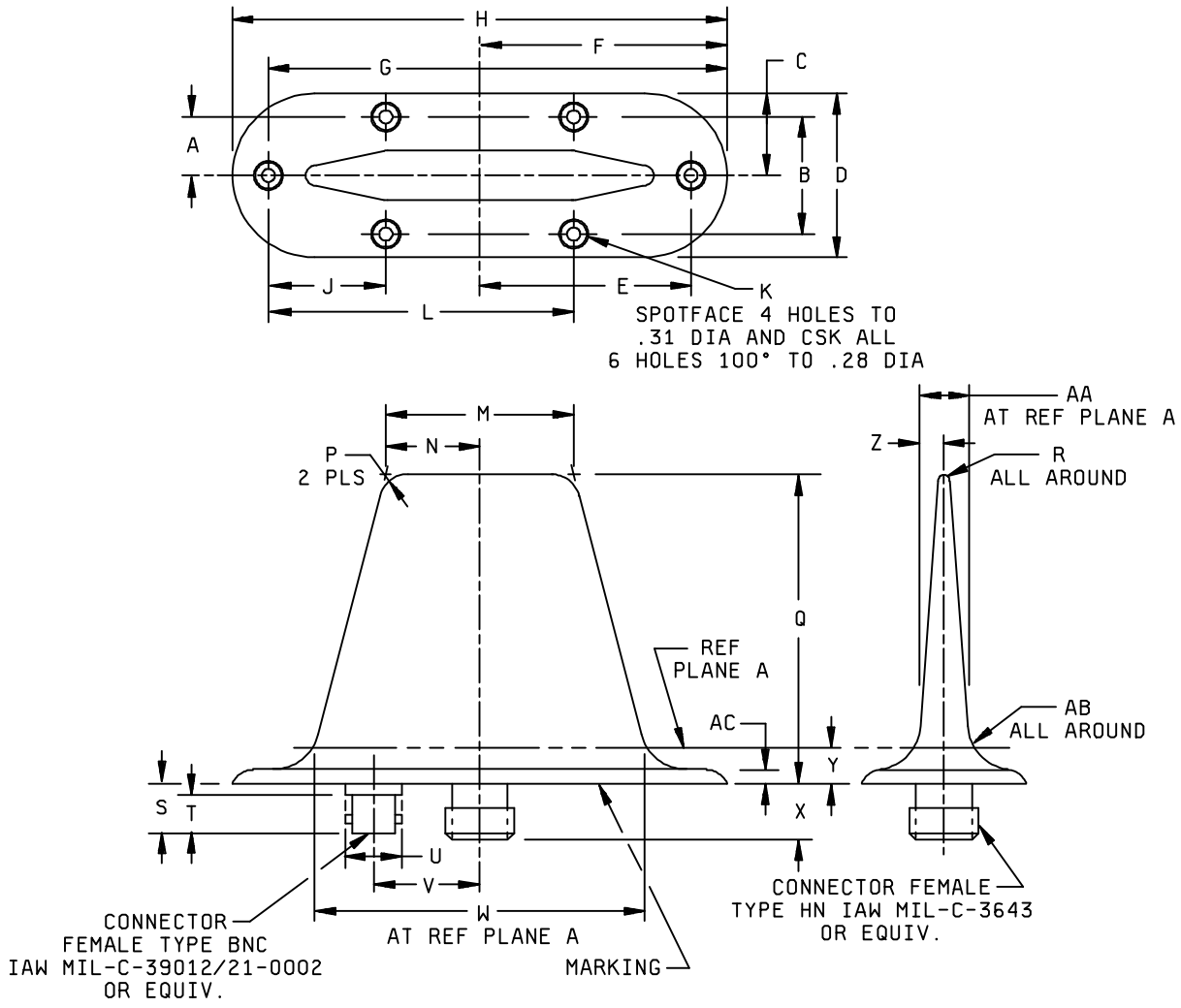



FIGURE 1. Outline drawing antenna AT-741B/B.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | .620 | .630 | 15.95 | 16.00 |
| B | 1.245 | 1.255 | 31.62 | 31.88 |
| C | .86 | .90 | 21.8 | 22.9 |
| D | 1.73 | 1.77 | 43.9 | 45.0 |
| E | 2.245 | 2.255 | 57.02 | 57.28 |
| F | 2.60 | 2.64 | 66.0 | 67.1 |
| G | 4.495 | 4.505 | 114.17 | 114.43 |
| H | 5.23 | 5.27 | 132.8 | 133.9 |
| J | 1.245 | 1.255 | 31.62 | 31.88 |
| K | .143 DIA | .149 DIA | 3.63 DIA | 3.78 DIA |
| L | 3.245 | 3.255 | 82.42 | 82.68 |
| M | --- | 2.00 | --- | 50.8 |
| N | --- | 1.00 | --- | 25.4 |
| P | .29 R | .33 R | 7.4 R | 8.4 R |
| Q | 3.28 | 3.32 | 83.3 | 84.3 |
| R | .06 R | .09 R | 1.5 R | 2.3 R |
| S | --- | .534 | --- | 13.56 |
| T | .414 | --- | 10.52 | --- |
| U | --- | .598 | --- | 15.19 |
| V | 1.11 | 1.15 | 28.2 | 29.2 |
| W | --- | 3.51 | --- | 89.2 |
| X | --- | .60 | --- | 15.2 |
| Y | .38 | --- | 9.7 | --- |
| Z | --- | .266 | --- | 6.76 |
| AA | --- | .532 | --- | 13.51 |
| AB | .20 R | .80 R | 5.1 R | 20.3 R |
| AC | .12 | .15 | 3.1 | 3.8 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. The base of the antenna is considered that part that rises from the bottom of the antenna mounting surface to the reference plane "A". Must be not less than .350 inch (8.89 mm). The cover is that part above reference plane "A".

FIGURE 1. Outline drawing antenna AT-741B/B - Continued.



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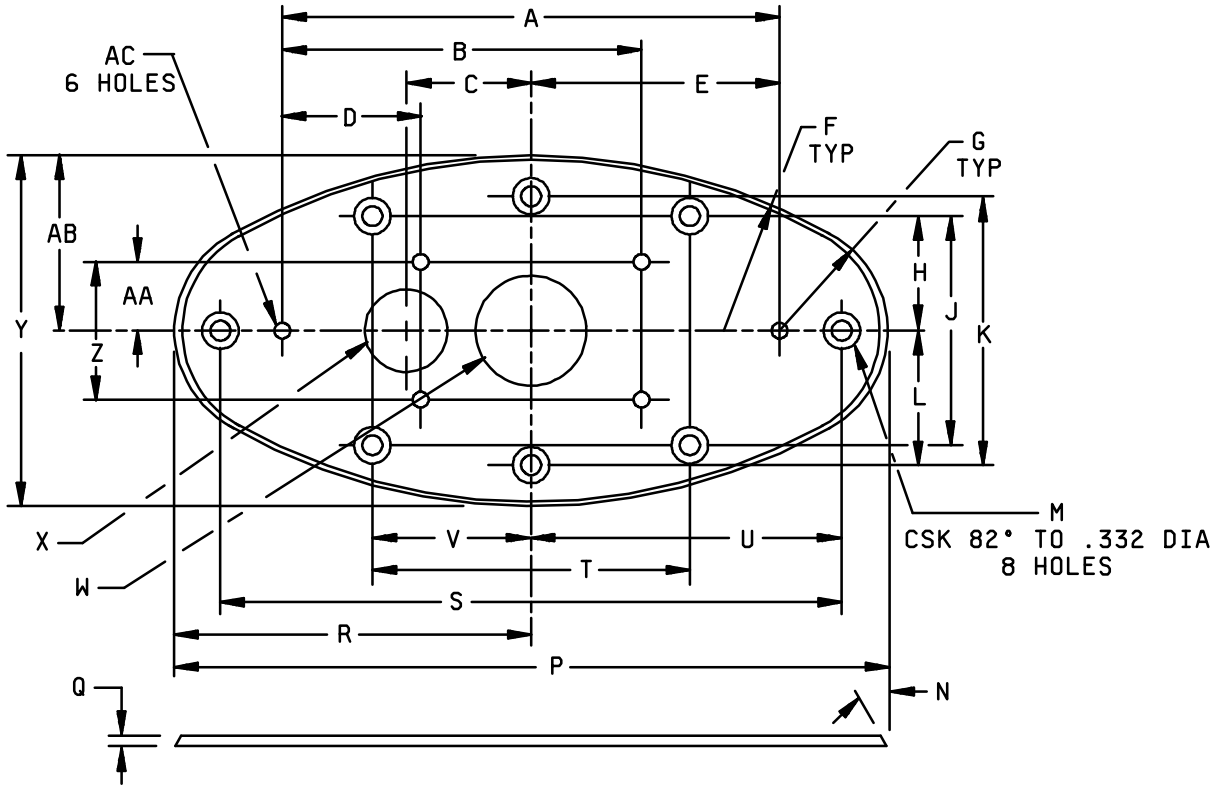



FIGURE 2. Adapter plate P/N M25708/1-02.



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| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | 4.495 | 4.505 | 114.17 | 114.43 |
| B | 3.245 | 3.255 | 82.42 | 82.68 |
| C | 1.12 | 1.14 | 28.5 | 29.0 |
| D | 1.245 | 1.255 | 31.62 | 31.88 |
| E | 2.245 | 2.255 | 57.02 | 57.28 |
| F | 5.49 R | 5.51 R | 139.5 R | 140.0 R |
| G | .99 R | 1.01 R | 25.1 R | 25.7 R |
| H | 1.034 | 1.044 | 26.26 | 26.52 |
| J | 2.073 | 2.083 | 52.65 | 52.91 |
| K | 2.432 | 2.442 | 61.77 | 62.03 |
| L | 1.213 | 1.223 | 30.81 | 31.06 |
| M | .179 DIA THRU | .185 DIA THRU | 4.55 DIA THRU | 4.70 DIA THRU |
| N | 25° | 35° | 25° | 35° |
| P | 6.43 | 6.45 | 163.3 | 163.8 |
| Q | .0902 | .0912 | 2.291 | 2.316 |
| R | 3.21 | 3.23 | 81.5 | 82.0 |
| S | 5.620 | 5.630 | 142.75 | 143.00 |
| T | 2.870 | 2.880 | 72.90 | 73.15 |
| U | 2.807 | 2.817 | 71.30 | 71.55 |
| V | 1.432 | 1.442 | 36.37 | 36.63 |
| W | .99 DIA | 1.01 DIA | 25.1 DIA | 25.7 DIA |
| X | .74 DIA | .76 DIA | 18.8 DIA | 19.3 DIA |
| Y | 3.18 | 3.20 | 80.8 | 81.3 |
| Z | 1.245 | 1.255 | 31.62 | 31.88 |
| AA | .620 | .630 | 15.75 | 16.00 |
| AB | 1.58 | 1.60 | 40.1 | 40.6 |
| AC | .143 DIA | .150 DIA | 3.63 DIA | 3.81 DIA |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 2. Adapter plate P/N M25708/1-02 - Continued.



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REQUIREMENTS (ANTENNA):

Dimensions and configuration: See figure 1.

Materials:

Base: Aluminum alloy in accordance with ASTM B85, alloy SC84B.

Cover: Suitable material that will resist erosion of impinging particles at velocities up to Mach 2.

Temperature operating range: -55°C to 250°C.

Static load: 15 psig.

Frequency range: 0.960 to 1.220 GHz.

Impedance: 50 ohms nominal.

VSWR:

Antenna: 1.40:1 maximum from 0.960 to 1.220 GHz.
1.30:1 maximum from 1.000 to 1.100 GHz.

Sampling probe: 1.50:1 maximum from 0.960 to 1.220 GHz.

Power: 4 kW peak, 100 watts average.

Weight: 10 ounces maximum.

Part or Identifying Number (PIN): See table I.

Data list: A data list shall accompany each antenna with VSWR measurement values made at 0.960, 1.000, 1.100, and 1.200 GHz.

REQUIREMENTS (ADAPTER PLATE):

Dimensions and configuration: See figure 2.

Material: Aluminum alloy in accordance with AMS-QQ-A-250, Temp T4.

Finish: Anodize in accordance with MIL-A-8625, Type I, Class I.

TABLE I. PIN.

| PIN | Description |
|---|---|
| M25708/1-01 | AT-741B/B Antenna |
| M25708/1-02 1/ | AT-741B/B Adapter |

1/ Inactive for new design. To be used to replace existing AS-133/APX and AT-722/APX antennas by using antenna AT-741B/B (M25708/1-01) and antenna adapter plate (M25708/1-02).

TABLE II. Supersession data.

| PIN | Replaces |
|---|---|
| M25708/1-01 | AT-741B/A |
| M25708/1-01 | AT-741A/A |
| M25708/1-01 | AT-741/A |
| M25708/1-01 and M25708/1-02 | AS-133/APX |
| M25708/1-01 and M25708/1-02 | AT-722/APX |



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Custodians:
Army - CR
Navy - AS
Air Force - 11

Review activities:
Army - AV
Air Force - 70, 71, 80, 84, 99

Preparing activity:
DLA - CC

(Project 5985-1158-001)



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